

Multiple Choice (10 marks)

1. Write the ratio of 96 runners to 216 swimmers in simplest form.
 - (a) 96 over 216
 - (b) 16 over 36
 - (c) 4 over 9
 - (d) 9 over 4
2. Jess scored 18 points during her last basketball game. Each basket she made was worth 2 points. How many baskets did she make?
 - (a) 20
 - (b) 16
 - (c) 9
 - (d) 8
3. Find the number that makes the statement true: $48 \text{ m} = \text{Box mm}$.
 - (a) 48,000
 - (b) 480
 - (c) 4,800
 - (d) 5
4. Which number written in exponential notation is equivalent to 343?
 - (a) 3^7
 - (b) 7^3
 - (c) 7^{49}
 - (d) 294^{49}
5. Emil wants to find out the most popular football team at a game between the home team and the visiting team. Which of the following methods will give him the most accurate results?
 - (a) surveying the cheerleaders for the home team
 - (b) surveying people wearing hats for the visiting team
 - (c) surveying a group of people standing in line for tickets
 - (d) surveying people who do not live in the home team's city

True / False (10 marks)

Answer True or False

6. True or False: The simplified form of $\frac{2+2i}{-3+4i}$ is $\frac{2}{25} - \frac{14}{25}i$.
7. True or False: The expression $(2 \times 4) \times 6$ is equivalent to 8×6 .
8. True or False: The solution to the equation gives x as $\frac{3}{4}$, indicating a common misinterpretation of the expression.
9. True or False: The object reverses direction two times in the interval $0 \leq t \leq 10$.
10. True or False: The product of 32 and 67 is 2,144.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A child has a set of 96 distinct blocks. Each block is one of 2 materials (plastic, wood), 3 sizes (small, medium, large), 4 colors (blue, green, red, yellow), and 4 shapes (circle, hexagon, square, triangle). How many blocks in the set differ from the 'plastic medium red circle' in exactly 2 ways? (The 'wood medium red square' is such a block) (A) 29 (B) 39 (C) 48 (D) 56 (E) 62
12. Discuss the relationship between the number of pears and apples Simon used in his fruit salad. Calculate the ratio of pears to apples and explain the significance of this ratio in the context of the fruit salad.

End of Paper